MILITARY SPECIFICATION

DC TO DC INVERTER, ELECTRONIC

This Amendment forms a part of Military Specification MIL-D-63144A (AR) dated 13 February 1981, and is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Insert "(M122)" following the designation "XM122." Make this addition wherever the designation "XM122" appears throughout this specification.

PAGE 1

2.1: Add the following specification under:

"SPECIFICATIONS - MILITARY"

"MIL-S-45743 - Soldering, Manual Type, High Reliability. Electrical and Electronic Equipment

MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies

MIL-P-55110 - Printed Wiring Boards"

PAGE 6

3.6.20: Add new paragraph as follows:

Temperature conditioning. The potted module shall be temperature soaked at -25°F+0°F-5°F for a minimum of three (3) hours prior to performing the test as specified in 3.6.2.

3.12: Add new paragraph as follows:

"Printed Wiring Board. The printed wiring board shall meet the requirements of MIL-P-55110 when tested in accordance with 4.4.3.1."

FSC 1375

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3.13: Add new paragraph as follows:

"Solderability of components. All components shall comply with the solderability requirement as specified in paragraph 3.4.3 of MIL-S-46844."

3.14: Add new paragraph as follows:

"Soldering operations. The assembly shall comply with the requirements of MIL-S-46844 for flow solder operations and MIL-S-45743 for hand soldering operations."

3.15: Add new paragraph as follows:

Printed wiring assembly cleaning. The assembly shall comply with all the requirements for cleaning after solder operations. Compliance shall be in accordance with MIL-S-46844 or MIL-S-45743 as applicable.

3.16: Add new paragraph as follows:

"Rework. All rework procedures shall be approved by the technical agency as noted in paragraph 6.4."

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4.4.2.1: Delete in its entirety and substitute new paragraph 4.4.2.1 as attached.

PAGE 14

4.4.2.2: Delete page 2 of 2 in its entirety and substitute page 2 of 2 as attached.

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4.4.3.2: Add new paragraph as follows:

"Solderability of components. Eighty (80) components (of all types used in the assembly) shall be randomly selected from each lot prior to production assembly and tested for the specified requirements in accordance with MIL-S-46844. If two (2) or more components fail the test criteria, the lot shall be classed defective. (Non-destructive test).

4.5.1.20: Add new paragraph as follows:

"4.5.1.20: Temperature conditioning. The module shall be temperature soaked to the specified requirements before performing the test as specified in 4.5.1.2."

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4.5.5: Add new paragraph as follows:

"Rework or retest. Any assembly failing to meet all of the requirements and is reworked or repaired shall be subject to all of the criteria of 4.4.2.1. Any assembly failing to comply with all of the requirements after rework shall be classed defective. All rework or repair shall be documented as to cause of rejection and the work performed."

4.5.6: Add new paragraph as follows:

"Rework or retest. Any assembly failing to meet all of the requirements and is reworked or repaired shall be subject to all of the test criteria of 4.4.2.2 before being classed acceptable. Any assembly failing to comply with all of the requirements after rework shall be classed defective. All rework or repair shall be documented as to cause of rejection and the work performed. (Non-destructive test)."

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6.3: Delete "DRDAR-QAA-I" and substitute "DRDAR-QAT-I."

6.4: Delete "DRDAR-QAA-M" and substitute "DRDAR-QAT-M."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A228)

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CLASSIFICATION OF DEFECTS & TESTS

AMENDMENT 1
MIL-D-63144A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	DC to DC Interter/Electronic Timer (Before Potting)				9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Components incorrectly installed, broken or missing		0.40%	3.2	Visual
102	Printed circuit board damaged		100%	3.2	Visual
103	Printed wiring board		--	3.12	4.4.3.1/MIL-P-55110
104	Solderability of components		--	3.13	4.4.3.2/MIL-S-46844
105	Soldering operations		100%	3.14	MIL-S-45743/MIL-S-46844
106	Printed wiring assembly cleaning		100%	3.15	MIL-S-45743/MIL-S-46844
107	Rework		100%	3.16	4.5.5
MINOR					
201	Evicence of poor workmanship		100%	3.11	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Inverter, DC to DC, Electronic Timer Module				9312632
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
126	Charging characteristics of the +55 volt source		100%	3.6.10	4.5.1.10
127	Rework		100%	3.16	4.5.6
128	Temperature conditioning		100%	3.6.20	4.5.1.20
MINOP					
201	Terminals missing or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		100%	3.11	Visual

NOTES:

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